

Problem Setup

September 23, 2026

Problem Setup

We are asked to solve the quadratic equation

$$x^2 + 3x + 0 = 1.$$

The given information is the equation itself; we need to find all real (or complex) values of x that satisfy it.

Strategy and Key Principles

The equation is a quadratic of the form $ax^2 + bx + c = 0$ after moving all terms to one side. We will use the **quadratic formula**

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a},$$

which provides the roots of any quadratic equation with coefficients a , b , and c . This method is appropriate because the equation cannot be factored easily over the integers.

Detailed Solution

$$x^2 + 3x + 0 = 1$$

$$x^2 + 3x = 1$$

$$x^2 + 3x - 1 = 0 \quad (\text{bring all terms to the left})$$

$$a = 1, \ b = 3, \ c = -1$$

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-3 \pm \sqrt{3^2 - 4 \cdot 1 \cdot (-1)}}{2 \cdot 1} \\ &= \frac{-3 \pm \sqrt{9 + 4}}{2} \\ &= \frac{-3 \pm \sqrt{13}}{2}. \end{aligned}$$

Final Answer

$$x = \frac{-3 \pm \sqrt{13}}{2}$$

Key Insights

- Always rewrite the equation in the standard quadratic form $ax^2 + bx + c = 0$ before applying the quadratic formula.
- The discriminant $b^2 - 4ac$ determines the nature of the roots; here it is positive (13), giving two distinct real solutions.
- A common mistake is to forget to move the constant term to the left side, which would lead to an incorrect discriminant.